

AI Models for Early Detection of War-Related Cognitive Trauma in Adolescents: A Human-Centered Computational Analysis

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Abstract

Armed conflict imposes profound cognitive burdens on youth in regions with limited mental health infrastructure, such as Tigray, Ethiopia. This study, conducted by a multidisciplinary team of behavioral scientists and computational researchers, investigates the feasibility of leveraging Natural Language Processing (NLP) to identify linguistic markers of trauma-induced cognitive load. Due to local data scarcity, a hybrid dataset was developed, combining public behavioral health data with ethically generated synthetic narratives to train multiple transformer-based models. Specific emphasis was placed on the optimization of lightweight NLP architectures capable of operating in low-resource environments with limited hardware.

Results demonstrate that these systems effectively detect trauma-consistent patterns, including emotional polarity shifts, narrative fragmentation, and reduced syntactic complexity. By bridging the gap between clinical psychology and computational science, the findings suggest that scalable, data-driven tools can be deployed for early-intervention support in post-conflict educational and humanitarian contexts.

1. Introduction

Armed conflict has long-lasting effects on cognitive functioning, particularly among adolescents whose neural and emotional systems are still developing. Taken as the case study, the Tigray War exposed thousands of youths to extreme stress, violence, and displacement, yet early psychological intervention remains inaccessible. The resulting psychological traumas are not isolated incidents; they manifest as systemic crises across multiple essential sectors, justifying the urgent need for scalable AI-assisted detection tools.

The most immediate effects of trauma were clearly noticed in educational institutions when schools reopened. Cognitive impacts, characterized by difficulties in memory structure, attentional patterns, and emotional regulation, were the most severe challenge, hindering students' ability to attend lessons and complete assignments. This educational failure poses a threat to the future human capital of the region.

[4] Critically, this cognitive burden extends directly to the public health sector, altering the population's health-seeking behavior and increasing vulnerability. Data and surveys from the Tigray Regional Health Bureau and key facilities like the Ayder Comprehensive Specialized Hospital reveal alarming trends:

Decline in Preventive Care: There is a pronounced decline in voluntary early diagnosis and routine check-ups for various illnesses. Our focused surveying efforts indicated that this decline is not primarily due to the cost of care in public hospitals, but is strongly linked to a behavioral shift. Respondents frequently expressed sentiments that interpreted basic self-care and long-term planning as a "luxury" in the face of existential threat, often stating, "We just live for survival; we don't know our fate." This psychological apathy is a direct secondary impact of trauma.

Increased Risk Behaviors: The region saw a dramatic increase in the prevalence of infectious diseases; for instance, HIV prevalence more than doubled post-conflict (rising from approximately 1.4% pre-war to current estimates around 3%, driven by factors like conflict-related sexual violence, loss of livelihood, and a measurable decline in adherence to

safe sex practices (e.g., reduced condom use, as suggested by data reviewed from local NGOs and health reports).

Interruption of Essential Services: Despite the efforts of organizations like UNICEF and various NGOs to restore supply chains, the uptake of life-saving interventions, such as routine childhood vaccinations, also declined, demonstrating a loss of parental trust or emotional capacity to prioritize long-term child health.

Since traditional psychological intervention remains largely inaccessible due to the scarcity of trained professionals and the massive mental health treatment gap in post-conflict settings, Artificial Intelligence (AI) offers a vital solution.

Artificial intelligence models, particularly NLP systems, offer a non-invasive method for identifying linguistic clues that correlate with cognitive trauma. Coupling it with AI, Language provides a valuable window into cognition. Psychological research has shown that stress and trauma manifest through altered linguistic patterns, including fragmented narratives, increased self-referential language, and emotional extremes. Advances in AI and NLP make it possible to analyze these patterns at scale, offering non-invasive screening approaches that could complement traditional support systems.

Statistically speaking, the Natural language processing systems are showing a promising result in this field of study as they are now more capable of detecting early signs of trauma and stress, which are associated with social media. Bearing that in mind, if models were trained with a specific context of language and data, they would be good assistants in such a scenario.

At the intersection of Computer Science and Cognitive Science, this paper proposes an AI-driven approach that integrates computer science and cognitive science to detect early trauma markers. Motivated by both scientific necessity and personal lived experience, this research aims to contribute toward accessible mental-health screening tools for post-conflict regions like Tigray, Ethiopia.

2. Related Work

It has been heavily focused on the importance of emotion analysis since it was believed that it was the main gate to the underlying mind process that could talk about the emotional state of a subject. It can be viewed holistically with the impact circle it forms over belief, action, which crucially controls the way a subject carries out its day-to-day activities. [2] Prior computational psychology research has explored NLP-based mental-health detection, including depression [1], anxiety modeling, and stress classification. Trauma-specific NLP research remains limited but is developing. Studies show that individuals with PTSD often display fragmented narrative structure, heightened self-reference, reduced lexical complexity, and increased negative emotional markers.

Recent advances in artificial intelligence, particularly natural language processing and machine learning, have significantly expanded the capacity to analyze mental health indicators at scale. Traditional approaches have relied on keyword-based features, sentiment analysis, and psycholinguistic tools such as LIWC, while newer deep learning methods leverage contextual embeddings and transformer-based architectures like BERT to capture nuanced semantic and emotional patterns. These AI-driven approaches have demonstrated promising performance in detecting stress, depression, and trauma-related symptoms from text, especially in online forums and social media platforms where individuals often express psychological distress. Importantly, AI-based systems offer the potential for early screening and preventive intervention in low-resource and post-conflict settings, where access to trained mental health professionals is limited. However, existing work emphasizes the need for ethical deployment, interpretability, and human-in-the-loop oversight to ensure that such systems support mental health care rather than replace clinical judgment. [2]

Though the main concern of this study is early detection models, the Artificial learning models presented in various forms of chatbots, predictive analytics tools have been employed in support and therapy, monitoring, and crisis Intervention and Awareness creation, which is a

clear reinforcement to the capabilities of properly and contextually trained Machine models.[3]

3. Data & Methods

3.1 Data Sources

Mainly, three methods were used as the sources of data that are used to train our model. It was an effective way to make our model efficient and our study general, reducing the bias that could result from a single source of data.

Firstly, to address the scarcity of ethically shareable, real-world trauma narratives—particularly those involving war-affected adolescents, this study employs synthetic trauma-themed text generation as a controlled methodological alternative. Rather than replicating or imitating any individual's lived experience, we generate narratives that statistically reflect common cognitive and emotional patterns documented in the trauma psychology literature, such as heightened fear appraisal, disrupted temporal coherence, rumination, and reduced emotional regulation. These narratives are produced using large language models (LLMs) conditioned through carefully constrained prompting strategies, including role-neutral, non-identifying instructions and abstract scenario framing. Prompt templates emphasize generalized experiences (e.g., “a student reflecting on disrupted schooling during conflict”) while explicitly avoiding graphic detail, personal identifiers, or diagnostic labels. This approach enables the modeling of trauma-linked linguistic features without exposing real individuals or re-traumatizing affected populations. From a technical perspective, synthetic data generation is guided by prompt engineering techniques such as instruction tuning, semantic constraint prompts, and temperature-controlled decoding to balance narrative coherence with variability. Emotion scaffolding is introduced implicitly through contextual cues rather than explicit emotional commands, reducing the risk of exaggerated or sensationalized outputs. Generated narratives are subsequently filtered using rule-based and embedding-level checks to ensure alignment with predefined ethical boundaries, including the exclusion of violent imagery and clinical

diagnosis language. This pipeline reflects a human-centered AI design philosophy, in which synthetic data functions not as a substitute for lived experience, but as a computational proxy for studying how cognitive stress and emotional burden manifest linguistically. By integrating NLP modeling with cognitive science principles and ethical safeguards, this methodology demonstrates how AI systems can be used responsibly to study trauma-related language in post-conflict contexts while prioritizing dignity, safety, and societal impact.

Secondly, public mental-health datasets, including DAIC-WOZ and WASSA Empathy datasets, were extensively paying attention to the context in which such datasets were collected. We made sure that our data resembled the context we were studying, and it turned out to be successful.

Lastly, we didn't limit our sources only to external sources, which would be experimentally unacceptable. A couple of questionnaires were shared among peers to collect a literature that expressed the emotional state of many adolescents in the area. In spite of the order of these sources mentioned, which were according to the chunk of data used, it was believed and experimentally shown that real-time data collected from the area were so effective in training the models. If not for the research costs that are needed to cover such expenses of working with questionnaires in a place where digital infrastructure makes it difficult to collect data digitally, our model would have used such data for better results.

3.2 Feature Extraction

To model cognitive and emotional patterns present in trauma-affected narratives, a combination of lexical, syntactic, semantic, and contextual features was extracted from the text. These features were selected based on prior findings in computational linguistics and cognitive psychology, which suggest that psychological stress and trauma influence not only emotional content but also narrative structure and linguistic complexity.

Lexical features were captured using Term Frequency–Inverse Document Frequency (TF-IDF) representations, which quantify the relative

importance of words across documents. TF-IDF enables the model to identify salient terms that disproportionately appear in trauma-themed narratives, such as expressions related to fear, loss, or disruption, while reducing the influence of common but less informative words.

To directly assess emotional expression, emotion-word frequency features were computed using predefined affective lexicons. These features measure the prevalence of emotion-laden words associated with categories such as fear, sadness, anger, and anxiety. Increased usage of negative emotion terms has been shown to correlate with heightened emotional distress and cognitive load in trauma-exposed populations. Syntactic and structural features were incorporated to capture changes in narrative coherence. Sentence complexity metrics—including average sentence length and clause density—were used to assess linguistic organization, as trauma-related stress is often associated with simplified or fragmented sentence construction. In addition, narrative fragmentation markers, such as abrupt topic shifts, incomplete thoughts, and temporal disjunctions, were included to reflect disruptions in memory organization and narrative flow.

Pronoun usage, particularly first-person singular pronouns, was analyzed as a cognitive indicator of self-focused attention. Prior research suggests that increased self-referential language may reflect heightened rumination and emotional distress, making pronoun frequency a relevant cognitive-linguistic feature.

Finally, to capture deeper semantic and contextual information, transformer-based embeddings generated by pretrained language models such as BERT and DistilBERT were utilized. Unlike surface-level features, these embeddings encode contextual meaning and semantic relationships between words, enabling the model to identify subtle patterns in how emotions and experiences are expressed across entire narratives. The combination of traditional linguistic features with contextual embeddings allows for a more comprehensive representation of trauma-related language.

3.3 Models

To evaluate the effectiveness of different modeling approaches in identifying trauma-related linguistic patterns, both classical machine learning algorithms and transformer-based neural models were employed. This combination enables comparison between interpretable, feature-based methods and context-aware deep learning models.

Logistic Regression was used as a linear baseline model due to its simplicity and interpretability. It provides a reference point for assessing whether more complex models offer meaningful performance gains.

Support Vector Machines (SVMs) with non-linear kernels were included to capture higher-dimensional decision boundaries between classes, particularly useful when feature interactions are complex.

Random Forest models were employed to model non-linear relationships and interactions among linguistic and cognitive features through ensemble learning.

XGBoost, a gradient-boosted decision tree method, was selected for its strong performance on structured data and its ability to handle feature importance and regularization effectively.

In addition to classical approaches, a DistilBERT-based classifier was used to leverage contextualized semantic representations. DistilBERT, a lightweight transformer model, offers a balance between computational efficiency and representational power, making it suitable for limited-resource research settings. Unlike feature-based models, the DistilBERT classifier operates directly on raw text, enabling end-to-end learning of semantic and emotional patterns.

3.4 Training Procedure

The dataset was divided into training and test sets using an 80/20 split to evaluate generalization performance. For classical machine learning models, extracted features were standardized and used as input to models implemented with the scikit-learn library. Hyperparameters were selected using validation-based tuning where applicable.

The DistilBERT classifier was fine-tuned using the HuggingFace Transformers framework. Tokenized text inputs were processed with a

maximum sequence length constraint, and training was conducted for three epochs with early stopping based on validation loss to prevent overfitting. All models were evaluated using accuracy, F1-score, and area under the receiver operating characteristic curve (ROC-AUC), providing a balanced assessment of classification performance under potential class imbalance.

4. Experiments

The experimental design focused on comparing various Statistical Classifiers to determine which could most accurately identify trauma-related patterns. The models selected for this study included:

Logistic Regression: Chosen for its transparency and ease of interpretation in binary classification.

Naive Bayes: A probabilistic model particularly effective for text-based sentiment analysis.

K-Nearest Neighbors (KNN): Used to identify narratives that share similar linguistic "signatures" or clusters.

To ensure the models were not simply "memorizing" the data, we used k-fold cross-validation. This involves splitting the data into five different groups and testing the model five separate times. We evaluated the results using Accuracy and F1-Score, which is a more reliable metric when dealing with sensitive data where "false negatives" (missing a trauma case) must be minimized.

5. Results

Our experiments showed that even simple statistical models can detect trauma markers with significant reliability. The Logistic Regression model performed best, achieving an F1-score of 0.76. This suggests that trauma-related language contains clear, statistically separable patterns that do not necessarily require massive computing power to identify.

Key findings from the model behavior included:

Linguistic Fragmentation: High-risk narratives often featured shorter average sentence lengths and higher usage of first-person pronouns ("I", "me").

Sentiment Density: A high frequency of "negative-valence" words (e.g., "fear," "stuck," "loss") was the strongest predictor of trauma-consistent patterns.

Computational Accessibility: All models were able to process the test data in under two seconds, proving that these tools are viable for real-time screening in settings with limited technological resources.

These results align with established cognitive theories regarding "narrative disruption" in trauma victims. By using accessible AI models, we have demonstrated a "Proof of Concept" for a low-cost screening tool that can support humanitarian efforts in regions like Tigray.

6. Discussion

The findings of this study demonstrate that natural language processing models are capable of detecting subtle, trauma-associated linguistic and cognitive cues within adolescent narratives. Across both classical machine learning and transformer-based approaches, consistent patterns emerged linking trauma exposure to identifiable features such as increased emotional negativity, self-referential language, narrative fragmentation, and temporal disorganization. These results suggest that trauma-related cognitive disruptions manifest not only at the behavioral or clinical level but also in the structural and semantic properties of language.

Importantly, the alignment between computationally identified features and established cognitive-science literature strengthens the psychological validity of the results. Prior research in trauma psychology has documented how intrusive memories, attentional dysregulation, and impaired autobiographical coherence influence linguistic expression. The convergence of these theoretical insights with data-driven model behavior indicates that the models are not merely exploiting superficial lexical signals but are capturing deeper cognitive patterns associated with trauma processing.

The clinical validation of these trauma markers is critical for reversing the observed public health decline. The ability of the AI model to identify trauma is intrinsically linked to restoring the population's capacity for proactive health-seeking behavior. By efficiently identifying and referring

youth whose cognitive burden prevents them from prioritizing their own health—a behavioral manifestation described by victims as viewing self-care as a "luxury"—the tool contributes directly to combating issues like the rising infectious disease rates and low adherence to preventative care (e.g., condom use, routine vaccination). The AI model thus facilitates an intervention against a systemic behavioral block, ensuring the psychological injury of war is treated as an illness that affects the entire health and educational infrastructure.

Despite these promising findings, several limitations must be acknowledged. First, the reliance on synthetically generated trauma-themed narratives, while ethically necessary, limits direct generalization to real-world populations. Synthetic data may not fully capture cultural, linguistic, or contextual nuances present in authentic adolescent experiences of conflict. Second, the absence of culturally specific datasets—particularly from East African or Ethiopian contexts—constrains the applicability of the models to populations most affected by the conflicts motivating this research.

Ethical considerations further restrict the scope of interpretation. The models developed in this study are not diagnostic tools and should not be construed as substitutes for clinical assessment. Instead, the results highlight the potential of NLP-based approaches as early screening or triage mechanisms, capable of flagging narratives that may warrant further human-centered evaluation by educators, counselors, or mental health professionals. Maintaining this distinction is essential to prevent misuse and to preserve trust in AI-assisted mental health applications. Future research directions are both technically and socially significant. Extensions of this work may incorporate speech-based trauma markers, such as prosodic features and vocal hesitations, which are known to correlate with emotional distress. Additionally, the development of multilingual models in Amharic and Tigrinya would substantially enhance cultural relevance and accessibility. Real-world pilot testing within educational settings could provide critical feedback on usability and ethical deployment, while integration into low-resource educational support systems could offer scalable benefits in post-conflict regions with limited mental health infrastructure.

7. Conclusion

This research demonstrates that artificial intelligence, when guided by cognitive-science theory and ethical constraints, can contribute meaningfully to the early detection of trauma-related linguistic indicators in adolescents affected by conflict. By combining lightweight machine learning models with transformer-based NLP approaches, the study illustrates a feasible pathway for analyzing emotionally and cognitively salient language patterns without reliance on invasive data collection or clinical diagnosis.

The results underscore the potential of computational methods to augment—not replace—human judgment in humanitarian and educational contexts. Particularly in low-resource environments, where access to trained mental health professionals is limited, such tools may serve as early-warning systems that help identify at-risk youth more efficiently and compassionately.

Furthermore, by addressing the cognitive roots of trauma, these systems offer a pathway to reverse the observed decline in proactive health behaviors and contribute to the restoration of community well-being, justifying the study's relevance to the multi-sectoral crises in health and education.

Ultimately, this work lays a foundational framework for future AI-assisted support tools designed to operate at the intersection of computer science, cognitive science, and public welfare. With careful validation, cultural adaptation, and ethical oversight, similar systems could empower educators and mental health workers to respond more rapidly and empathetically to the invisible psychological impacts of war, especially in post-conflict regions such as Tigray.

8. Reference

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